Security Operations Center (SOC) Alert Management Report

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➢ Introduction

This report outlines the implementation of an alert management workflow using SIEM tools and structured methodologies. The objective is to classify, prioritize, and respond to security alerts efficiently while aligning with industry frameworks such as MITRE ATT&CK and CVSS scoring.

➢ Alert Classification System

An alert classification framework was developed using structured fields including alert type, severity, and MITRE ATT&CK technique mapping.

Alert ID	Type	Priority	MITRE Technique	Description
001	Phishing	High	T1566	Suspicious email with malicious link
002	SSH Brute Force	Medium	T1110	Multiple failed login attempts
003	Ransomware	Critical	T1486	File encryption detected
004	Port Scan	Low	T1046	Multiple ports scanned
005	Command Exec	High	T1059	Suspicious shell command

This classification enables consistent alert triage and faster incident handling.

➢ Alert Prioritization Using CVSS

Alerts were prioritized based on CVSS scores to determine severity and response urgency.

Alert Name	CVSS Score	Priority
Log4Shell	9.8	Critical
Ransomware	9.5	Critical
Phishing Email	7.2	High
SSH Brute Force	6.5	Medium
Port Scan	3.5	Low

Severity Mapping Logic:
Critical: 9.0 – 10
High: 7.0 – 8.9
Medium: 4.0 – 6.9
Low: 0 – 3.9

This approach ensures that high-risk vulnerabilities receive immediate attention.

➢ Dashboard Visualization (Wazuh)

A dashboard was created in Wazuh to visualize alert distribution and trends.

Key Visualizations:
Alerts by Severity (Pie Chart)
Top Alert Types (Bar Chart)
Alert Volume Over Time (Line Graph)

Configuration Highlights:
Alerts grouped by priority levels
Color coding applied:
Critical → Red
High → Orange
Medium → Yellow
Low → Blue
This visualization improves situational awareness and supports quick decision-making.

➢ Incident Ticket Creation (TheHive)

A sample incident ticket was created to document and track a critical security event.

Title:
[Critical] Ransomware Detected on Server-X

Description:
A ransomware infection was identified on Server-X, indicating active file encryption behavior.

Indicators of Compromise (IOCs):
File: crypto_locker.exe
IP Address: 192.168.1.50

Priority: Critical
Status: Open
Assigned To: SOC Analyst (Tier 1)

Recommended Actions:
Isolate the affected system
Block malicious IP address
Perform forensic investigation
Escalate to Tier 2

➢ Threat Intelligence Validation

An IP address (192.168.29.117) was analyzed using threat intelligence tools.

Findings:
Community Score: 0/94
No security vendors flagged the IP as malicious
Classified as a private/internal IP

Analysis:
The IP appears benign and likely part of an internal network. No immediate threat indicators were identified, but correlation with SIEM logs is recommended.

Escalation Procedure
A structured escalation process ensures timely response to critical incidents. (Document attached)

➢ Conclusion

This practical implementation demonstrates a complete SOC workflow, including alert classification, prioritization, visualization, incident management, and escalation.

The integration of structured frameworks such as MITRE ATT&CK and CVSS enhances detection accuracy and response efficiency. The use of SIEM tools like Wazuh and case management platforms like TheHive ensures streamlined security operations.